

Appendix S6. Shrew taxonomy

Our optimal topology (Fig. 4) supports Soricidae, the crown group encompassing all descendants of the last common ancestor of living shrews, with a posterior probability of 1.0. Soricinae also exhibits a posterior probability of 1.0. We agree with Hutterer (2005) that a third clade, Myosoricini, is monophyletic, here reconstructed within Crocidurinae. Given our phylogenetic sample of *Myosorex* and *Congosorex*, infrafamilial (Myosoricinei) and tribal (Myosoricini) ranks for this clade are synonymous. However, distinct ranks could be applied in future studies that also sample the third genus (*Surdisorex*). We use the common noun suffix "ines" to indicate subfamily (e.g., crocidurines for Crocidurinae). Due to the potential confusion of the suffix "ins" with infrafamily or tribe, we avoid using common nouns at those ranks in favor of the proper Linnean noun, e.g., "species of Myosoricini or Crocidurinei" rather than "myosoricins or crocidurins". We place Crocidurinei and Myosoricinei within Crocidurinae to maximize continuity with previous usage (see Table below and Repenning 1967: fig. 42). Due to the variable placement of several fossils, Myosoricinei and Crocidurinae are not as strongly supported in our timetree, but have posterior probabilities of 1.0 in our analysis of living taxa alone (Fig. S1a). We use Soricoidea for the total group including Soricidae and Heterosoricidae. In our unconstrained, non-clock Bayesian analysis (Fig. S8a), Soricoidea and Heterosoricidae received posterior probabilities of 1.0 and 0.99, respectively. Heterosoricid monophyly in our timetree (Fig. 4) supports Reumer's (1998:12) classification of the taxa known to him at the time, including *Domnina*, *Quercysorex*, *Ingentisorex*, *Dinosorex*, and *Heterosorex*. The then-unknown *Lusorex* was, in our view, correctly identified as a heterosoricid by Storch & Qiu (2004). Our results suggest that *Domnina* and *Quercysorex* may be paraphyletic at genus level (Fig. 4) but most of their constituent species are not yet sufficiently well-known to fully test this possibility. Based on our timetree (Fig. 4), Heterosoricidae exhibits synapomorphies including a bifid (or "fissident") apex of the upper incisor and a postsymphyseal foramen.

Soricidae includes several extinct lineages of "crocidosoricines", which we reconstruct as closer to crocidurines than soricines, albeit without strong support. Indeed, in the non-clock Bayesian analysis, *Shargainosorex* (Zazhigin & Voyta 2018) appears as the sole member of stem Soricinae, with the remaining "crocidosoricines" forming a polytomy at the base of Soricidae (Fig. S8a). Parsimony with implied weighting (Fig. S8b) also places *Shargainosorex* at the base of soricines, and weakly resolves some "crocidosoricines" within crocidurines (e.g., *Lartetium*) and others outside of crown Soricidae (e.g., *Oligosorex*).

As in some previous phylogenies with an adequate sample of soricids (Dubey et al. 2007; Willows-Munro et al. 2011; Faurby & Svenning 2015), we reconstruct the genus *Suncus* as paraphyletic. Our Bayesian analyses (Figs. 4, S1a, S8a) reconstruct *S. murinus* closer to *Crocidura*-*Diplomesodon* than to "*S.*" *etruscus*. *Paradoxon* Wagner 1855 may be the most appropriate genus name for "*Suncus*" *etruscus* (Hutterer 2005), formal revision of which must await an analysis including more of the ca. 18 species traditionally assigned to this genus. Similarly, the position of "*Diplomesodon*" within the genus *Crocidura* renders the latter paraphyletic, leading us to recommend the binomen *Crocidura* ("*Diplomesodon*") *pulchellum* in order to retain the validity of this widely-known mammalian genus (Hutterer 2005).

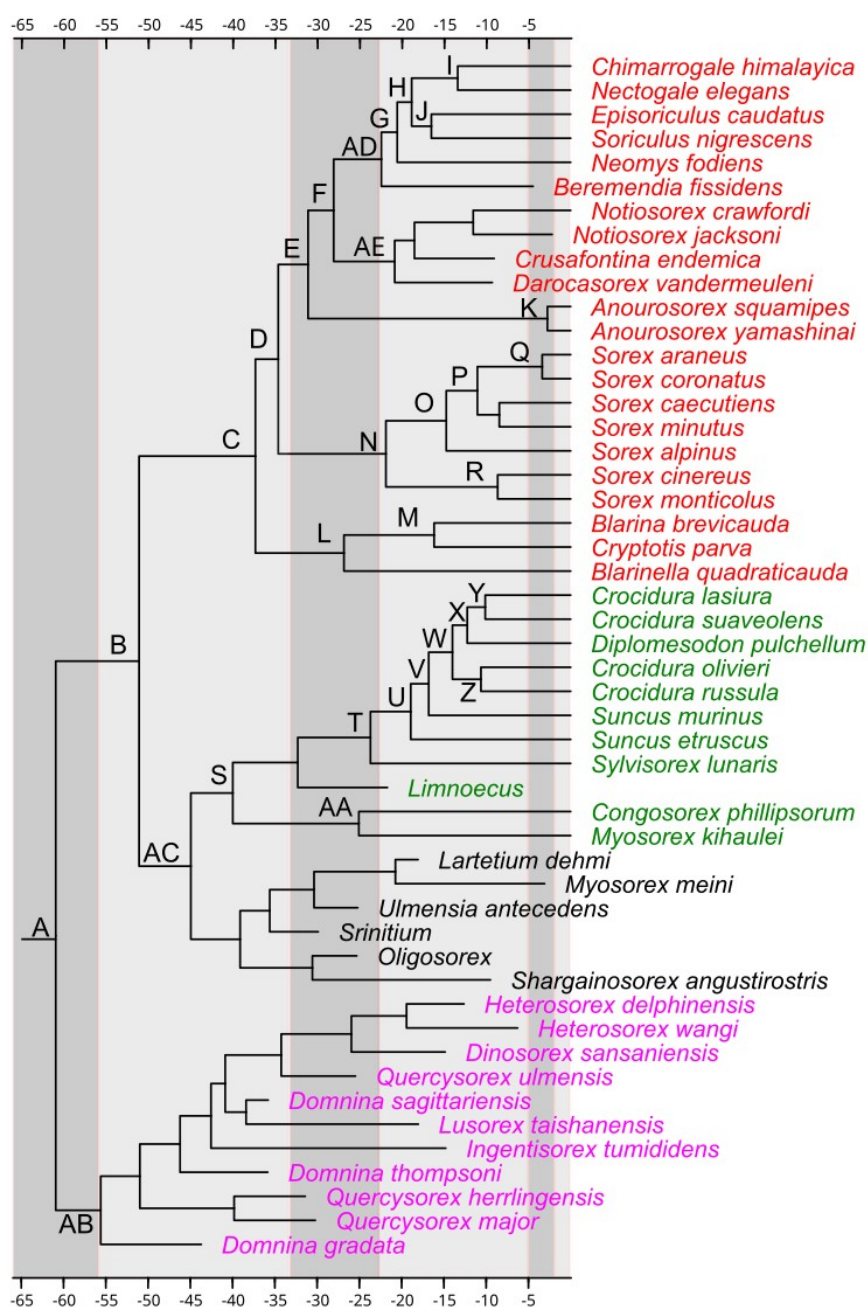
According to our timetree (Fig. 4), no fossil in our sample represents a stem soricine. Crown Soricinae shows an initial divergence between Blarininei and Soricinei, followed by *Sorex* as sister to all other soricines in Neomyini. We reconstruct *Anourosorex* closer to Neomyina (see table below) than *Sorex* or Blarininei. The European Neogene fossils *Beremendia* (Reumer 1984), *Crusafontina* (vanDam 2004), and *Darocasorex* (vanDam 2010), as well as the North American species *Notiosorex jacksoni* (Hibbard 1953), all appear within crown Soricinae, consistent with the paleontological literature (op cit. and Furió et al. 2007, 2010). Within soricines, we reconstruct *Darocasorex* and *Crusafontina* closer to *Notiosorex* than *Anourosorex*, contra vanDam (2010), and *Beremendia* closer to *Nectogalita* than Blarininei, contra Furió et al. (2010).

Taxonomy of Soricoidea derived from our optimal timetree (Fig. 4). Suffixes follow McKenna & Bell (1997:513), in turn based on the International Code of Zoological Nomenclature, article 29. High level taxa are provided for clades we recover as monophyletic and derive from genus-level types implicit in their root names. Where possible, we follow the taxonomy of Repenning (1967) and Hutterer (2005). "Total" or "crown" indicate applicability of name to total or crown group, respectively. Species membership in genus names between quotes should change to maximize monophyly. Letters after high-level taxa correspond to nodes on timetree shown below; "A" corresponds to Soricoidea. Not all nodes are named here but still have an age estimate; see Appendix S9, "Summary of divergence time estimates".

Family	Subfamily	Infrafamily	Tribe	Subtribe	Infratribe	Genus
Heterosoricidae (AB)						<i>Heterosorex</i>
						<i>Dinosorex</i>
						<i>Ingentisorex</i>
						<i>Lusorex</i>
						" <i>Quercysorex</i> "
						" <i>Domnina</i> "
Soricidae (B)	Crocidurinae total (AC)					<i>Srinitium</i>
						<i>Ulmensia</i>
						<i>Oligosorex</i>
						<i>Lartetium</i>
						<i>Shargainosorex</i>
					" <i>Myosorex</i> " <i>meini</i>	
	Crocidurinae crown (S)					<i>Limnoecus</i>
Myosoricinei (AA)		Myosoricini (AA)			<i>Myosorex</i>	
					<i>Congosorex</i>	
Crocidurinei crown (T)					<i>Sylvisorex</i>	
						" <i>Suncus</i> "
			Crocidurini (U)			<i>Crocidura</i>
						" <i>Diplomesodon</i> "
	Soricinae (C)	Blarininei (L)	Blarinini (M)			<i>Blarina</i>
						<i>Cryptotis</i>
			Blarinellini			<i>Blarinella</i>
		Soricinei (D)	Soricini (N)			<i>Sorex</i>
				Neomyini (E)	Anourosoricina (K)	
				Neomyina (F)	Notiosoricita total (AE)	<i>Crusafontina</i>
						<i>Darocasorex</i>
						<i>Notiosorex</i>
					Nectogalita total (AD), crown(G)	<i>Beremendia</i>
						<i>Neomys</i>

Family	Subfamily	Infrafamily	Tribe	Subtribe	Infratribe	Genus
					Nectogalita continued, total (AD), crown(G)	<i>Soriculus</i>
						<i>Episoriculus</i>
						<i>Nectogale</i>
						<i>Chimarrogale</i>

Image below is a guide to lettered taxonomic designations in the above table. Topology is a closeup of Soricoidea from the timetree shown in Fig. 4 (*Adunator* root calibration, narrow priors, as explained in Methods). Red indicates Soricinae, Green Crocidurinae, pink Heterosoricidae. Numbers at top and bottom indicate millions of years before present.



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